REPORT

PATIENT DATA PREPROCESSING AND SEGMENTATION

DATA PREPROCESSING AND VISUALIZATION'S PROJECT

PREPARED BY:
TEAM CITY HOSPITAL · DATA SCIENCE GROUP · DATA PREPROCESSING TEAM

TEAM MEMBERS:
CAROLINA AREZ, 20241739
HENRIQUE MADUREIRA, 20241756
MIGUEL MELO, 20241757
NOA PENAS, 20241682
GROUP 4

NOVAIMS

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Introduction

Hospitals must continuously improve patient care and operational efficiency while understanding patient needs and satisfaction. City Hospital, operating across multiple departments, seeks to leverage its information systems data to optimize services and decision-making.

Our team is the Data Preprocessing (DP) unit of City Hospital's Data Science group. Our primary responsibility is to transform raw patient data into a structured, analysis-ready format. This involves thorough data understanding, cleaning, and feature engineering, culminating in the construction of an Analytic Base Table (ABT) at the patient level, with one row per patient. The ABT consolidates static demographic attributes, aggregated behavioral metrics from all visits, satisfaction measures, and activity-related features.

Beyond data preparation, we performed exploratory analysis to give City Hospital management insights into patient demographics, department usage, consultation durations, and satisfaction trends

Overall, this report documents our preprocessing approach, the ABT construction, and the exploratory visualizations, demonstrating how structured data can drive evidence-based decisions and enhance patient-centered care at City Hospital.

Project Structure Overview

The work developed in this project is organized into **three main parts**, following a structured data science workflow:

Part	Focus	Outcome
Part A	Data Understanding and Cleaning	Explore the raw dataset and identify/resolve data quality issues, such as inconsistent values, missing data, duplicates, and outliers.
Part B	Feature Engineering and Analytic Base Table (ABT) Construction	Transform visit-level data into a consolidated patient-level dataset suitable for analytical tasks.
Part C	Exploratory Visualization and Insights	Analyze the ABT through visualizations to extract meaningful business insights about patient demographics, service usage, satisfaction, and financial behavior.

1.Part A – Data Understanding & Cleaning

The objective of Part A was to ensure that the raw dataset obtained from City Hospital's information systems was properly understood, cleaned, and prepared for further analytical tasks, including feature engineering, ABT construction, and exploratory analysis. This phase focused on identifying and correcting data quality issues such as inconsistent data types, duplicate records, illogical values, missing data, and outliers.

1.1 Data Import

The dataset was imported from an Excel file into Python using the Pandas library. A working copy of the original dataset was created to preserve the raw data and allow safe transformations during the cleaning process.

1.2 Initial Understanding of the Dataset

An initial exploration of the dataset was performed to understand its structure and content. The first rows of the dataset were inspected, and a general overview of the variables was obtained using summary information.

At this stage, the dataset contained **10,008 observations and 16 variables**, including:

- Patient demographics (Age, Gender, City, Profession, Education Level, Marital Status);
- Clinical variables (Department, Consultation Duration, Satisfaction Level);
- Financial variables (Approximate Annual Income, Consultation Price, Insurance Coverage);
- Administrative variables (Patient ID, Visit Date, Insurance Provider, Family History).

The visit dates ranged from **January 1st, 2024 to June 30th, 2024**, confirming a six-month observation window. All categorical variables were inspected to identify their unique values, ensuring that only valid categories were present.

Column	Possible Values
Gender	Female, Male, Other
City of Residence	Belfast, Birmingham, Dublin, Edinburgh, Glasgow, Liverpool, London, Manchester
Profession	Artist, Businessperson, Doctor, Engineer, Lawyer, Nurse, Retired, Scientist, Student, Teacher
Insurance Provider	Provider A, Provider B, Provider C, Provider D
Family History	Cancer, Diabetes, Heart Disease, Hypertension
Education Level	Elementary school, High school, Kindergarten, Master, Middle school/junior high, PhD, Pre-school, Undergraduate
Marital Status	Divorced, Married, Single, Widowed
Department	Allergology, Cardiology, Dermatology, ENT, Emergency, Endocrinology, Gastroenterology, General Practice, Neurology, Orthopedics, Psychiatry, Pulmonology, Rheumatology

Table 1 – Categorical Columns and Their Unique Entries

In addition, numerical columns were examined by checking their minimum and maximum values. This revealed several unrealistic values, such as:

- Age equal to 0 and as high as 195;
- Satisfaction levels reaching 6 instead of the expected 1–5 range;
- Extremely high values for income and consultation prices.

Column	Type	Range
Patient ID	int64	1000 → 1499
Age	float64	0.0 → 195.0
Visit Date	datetime64	2024-01-01 00:00:00 → 2024-06-30 00:00:00
Consultation Duration	int64	15 → 600
Satisfaction Level	int64	1 → 6
Approximate Annual Income	float64	0.0 → 11970900.0
Consultation Price	float64	50.03676194421384 → 39999.2220179375
Insurance Coverage	float64	0.0 → 421.887836229827
Total entries	None	10008

Table 2 – Numerical Columns and Their Unique Entries

These findings indicated the presence of outliers and logical inconsistencies that required further treatment.

1.3 Handling Duplicate Rows and Inconsistent Data Types

A verification of data types confirmed that most variables were stored in a consistent format. As a preventive measure, missing values encoded as text were explicitly converted into proper missing values (NaN) for selected categorical variables**.**

Duplicate rows were then identified across the dataset. Several pairs of fully duplicated records were detected, representing repeated consultation entries. These duplicate observations were manually removed to avoid double-counting and bias in later aggregation at the patient level.

1.4 Searching and Fixing Inconsistencies

To ensure internal consistency at the patient level, variables that should remain constant for the same patient were analysed. These included:

- Gender;
- City of Residence (no patient has multiple cities recorded);
- Profession;
- Insurance Provider (no patient has multiple providers recorded);
- Family History (no patient has multiple family history entries recorded);
- Education Level (no patient has multiple education levels recorded);

- Marital Status;
- Approximate Annual Income;
- Age.

By grouping records by **Patient ID** and counting the number of unique values per variable, several inconsistencies were detected:

- Some patients had conflicting values for **Gender**, appearing as both male and female.

Patient ID	Gender	Female	Male
1050	Male, Female	14	3
1307	Male, Female	1	23
1349	Male, Female	1	17
1447	Female, Male	18	2
1490	Female, Male	24	3

Table 3 – Patients with Conflicting Gender Records

To address this issue, we used the mode to determine the most frequently occurring gender in each patient's records. For example, for patient ID 1050 in the table above, there are 14 records labelled as Female and 3 records labelled as Male. Based on the mode, we concluded that the patient is Female, and the 3 records incorrectly labelled as Male were corrected to Female.

- **Profession** changed for some patients across visits:

Patient ID	Profession	Student	Retired
1488	Student, Retired	17	2
1496	Retired, Student	5	13

Table 4 – Patients with Profession Changes Across Visits

To resolve this issue, the same approach was applied. The mode of the Profession was computed across all records for each patient, and any incorrectly labelled entries were corrected to match this value.

- **Marital Status** showed inconsistent reporting:

Patient ID	Marital Status	Single	Widowed	Married
1140	Single, Widowed	18	1	0
1322	Single, Married	17	0	2
1332	Single, Married	23	0	1
1382	Single, Widowed	21	1	0

Table 5 – Patients with Profession Changes Across Visits

To correct this issue, we again used the mode. For each patient, we calculated the most frequently occurring Marital Status across all records and updated any mismatched entries accordingly.

- **Income** had multiple different values for the same patient:

Patient ID	Approximate Annual Income
1000	[54388. 35475. 33272. 38259. 32950. 52519. 45824. 55025. 46193. 30845. 33364. 31503. 56713. 33383. 53857. 57119. 51859. 45561. 49214. 36022.]
1001	[59833. 40834. 41843. 50418. 38074. 36327. 53120. 39036. 35901. 57443. 41987. 34736. 40155. 34675. 55302. 47747.]
1002	[0. nan]
1003	[53760. 44496. 39929. 39859. 32489. 40125. 53462. 45248. 39047. 41765. 36599. 35120. 47618. 31873. 45797. 52072. nan 51764. 31496. 34652. 45069. 50411. 49804. 46624. 34008.]
1005	[42941. 34834. 43009. 38020. 55643. 56549. 32508. 48594. 37703. 40046. 31485. 52015. nan 41773. 47771. 39396. 46436. 58774. 56285. 44802. 49718. 52573. 54111. 30392. 37054. 52042. 45372. 58196. 53541.]

Table 6 – Example of First 5 Patients with Income Changes Across Visits

To correct the income inconsistencies, a different approach was required because income is not only a numerical variable but also a continuous one, making the mode inappropriate in this context. Instead, we used the median, since outliers had not yet been treated and the median is far less sensitive to extreme values than the mean. For each patient, we computed the median income across their records and replaced all income entries for that patient with the corresponding median value.

- Only one patient exhibited **age** inconsistencies. We observed that all but one of the records listed the patient's age as 138, while a single record reported the age as 38. Since 138 is clearly unrealistic, we identified this as a typographical error and manually corrected all records stating 138 to 38, which is a much more reasonable age.

1.5 Filtering Non-Logical Occurrences

Logical validation was applied to identify unrealistic values in the variables, considering that the data are based on the UK context. We primarily examined seven major groups of potential illogical occurrences to determine whether any were present.

- Education Level vs Age:
  - **Master's degree** and **age < 21**. While there is no minimum age to obtain a master's degree, it is almost impossible to acquire one before turning 21 years old;
  - **PhD** and **age < 24**. While there is no minimum age to obtain a master's degree, it is almost impossible to acquire one before turning 24 years old;
  - **Undergraduate** and **age <18**. In practical and real-world UK education systems, obtaining a completed undergraduate degree before turning 18 is considered not realistically possible;
  - **Middle School** and **age < 11**. In the UK, finishing Middle School before age 11 is not realistically possible and is considered highly implausible in our dataset;
  - **Elementary School** and **age < 6**. It is not possible to finish elementary/primary school before age 6. Primary education typically runs from ages 5 to 11, so completing it under 6 would be impossible in practice;

- **Pre-school** and **age < 3**. In the UK, finishing pre-school before age 3 is not realistically possible;
- **Kindergarten** and **age < 3**. In the UK, finishing kindergarten before age 3 is not realistically possible.

- Profession vs Age:
  - **Not being a student** and being **under 18 years** of **age**. In the UK, a person may be under 18 and not be a school student; however, they are legally required to be in some form of education or training until the age of 18. Given that all the professions observed in our dataset other than 'student' require at least one degree, such cases are therefore considered illogical values;
  - **Doctor** and **age < 23**. In the UK, being a fully qualified doctor under the age of 23 is not realistically possible. Medical degrees typically take 5–6 years to complete, followed by 1–2 years of foundation training. Therefore, even if a student starts their medical degree at 18, they would only be able to finish and qualify as a doctor at around 23 years of age;
  - **Lawyer** and **age 23**. In the UK, becoming a fully qualified lawyer before the age of 23 is highly unlikely. Completing a law degree (3 years) or a conversion course (1 year), followed by the required professional training and practical experience (1–2 years for LPC/SQE and 1–2 years for training contract or pupillage), means that the earliest realistic age to qualify is around 23–24;
  - **Retired** and **age < 50.** In the UK, retiring before the age of 50 is highly unusual. The state pension age is currently over 50, and while early retirement is possible in some private or occupational schemes, it is rare. Therefore, seeing 'retired' with age under 50 in our UK-based data is considered highly implausible.

- Marital Status vs Age:
  - **Married** and **age < 16**. Married and age < 18. In most of the UK (England, Wales, Northern Ireland and Ireland), the minimum legal age to marry is 18. In Scotland, it is possible to marry at 16 with parental consent. Therefore, records indicating marriage under 16 are considered invalid or highly implausible, while those at 16–17 are uncommon and region-dependent;
  - **Divorced** and **age < 16**. Since it is not legally possible to marry before the age of 16 in the UK, it is therefore impossible to be divorced before turning 16;
  - **Widowed** and **age < 16.** Since it is not legally possible to marry before the age of 16 in the UK, it is therefore impossible to be widowed before turning 16;
- Education-Profession inconsistencies:
  - **Doctor without degree**. In the UK, becoming a doctor requires completing a medical degree;
  - **Lawyer without degree**. In the UK, becoming a lawyer requires completing a law degree or a conversion course;

  - **Scientist without degree**. In the UK, professional scientific roles typically require at least an undergraduate degree;
  - **Engineer without degree**. In the UK, professional scientific roles typically require at least an undergraduate degree;
  - **Nurse without degree**. In the UK, registered nurses are required to have a nursing degree or equivalent qualification;
  - **Teacher without degree**. In the UK, qualified teachers must have at least an undergraduate degree and a teaching qualification;
  - **Non-Student with education equal to Pre-school or Middle School**. Since all the professions in our dataset require at least an undergraduate degree, it is not possible to practice any of them with only pre-school or middle school education.

- Profession-related inconsistencies:
  - **Student and widowed**. While it is legally possible for a student to be widowed, such cases are extremely rare. Therefore, records indicating a student who is widowed would have been carefully verified for plausibility. (In this case, we found out there were no widowed students);
  - **Student and age > 65**. While lifelong learning is possible, it is highly unusual for someone over 65 to be a full-time student. Records showing a student older than 65 are considered highly unusual and were verified for accuracy.

- Satisfaction levels outside the valid 1–5 range. In the description of the dataset, we were told the Satisfaction Levels range from 1 to 5, so we checked if any records have values that fall of that range. To fix this, we changed all entries with Satisfaction Level equal to 6, to 5;

- Records with insurance coverage greater than consultation price.

After checking all these groups, we found the following inconsistencies:

- **353** records stated that the patient had **age < 3 and an Education Level of Pre-school**. In these cases, we **updated the education level to 'None',** since children under 3 in the UK do not necessarily attend formal educational institutions, some may attend nurseries, while others are cared for at home;

- **112** records stated that the patient was **not a student and had age < 18**. We updated all these patients' **professions to 'student'**, since, as noted above, it is not plausible for individuals under 18 to hold any of the professions listed in our dataset;

- **20** records stated that the patient was **younger than 23 years old and were doctors**. Since there were no other reasonable ways to correct these implausible records (excluding deleting them, which we chose to avoid), we updated the **profession to 'student'**;

- **94** records stated that the patient was **younger than 22 years old and were lawyers**. Since there were no other reasonable ways to correct these implausible records (excluding deleting them, which we chose to avoid), we updated the **profession to 'student'**;

- **32** records stated that the patient was retired and **younger than 50 years old**. We didn't do anything specific to fix these records. Somehow, they were fixed by other lines of code (we know this because after running all the fixing code, we checked again to see if it had worked);

- **20** records stated that the patient was **a doctor with no degree**. These records were automatically corrected when we addressed cases where patients under 18 were assigned any profession other than 'student,' so no additional coding was required to fix them specifically;

- **25** records stated that the patient was **a lawyer with no degree**. These records were corrected when we updated cases where patients under 22 were listed as lawyers, so no additional coding was required to fix them specifically;

- **35** records stated that the patient was **a teacher with no degree**. We didn't do anything specific to fix these records. Somehow, they were fixed by other lines of code (we know this because after running all the fixing code, we checked again to see if it had worked);

- **63** records stated that the patient was **not a student and the highest education level they got was Pre-School or Kindergarten**. We didn't do anything specific to fix these records. Somehow, they were fixed by other lines of code (we know this because after running all the fixing code, we checked again to see if it had worked);

- **27** records stated that the patient was **older than 65 years old and was a student**. All these records referred to a patient aged 118. Considering the patient's age, we updated the **profession from 'student' to 'retired';**

- **6** records stated that the **insurance coverage was greater than the consultation's price**. To correct this, we capped the insurance coverage at the consultation price for these records, ensuring that coverage did not exceed the consultation cost, which would be illogical.

1.6 Handling Outliers

Outliers were analysed for key numerical variables, including Age, Consultation Duration, Satisfaction Level, Approximate Annual Income, Consultation Price, and Insurance Coverage. Since some values in these variables had been modified earlier, we re-examined their ranges to better understand the data and identify any remaining anomalies (we also checked the range for the Patient ID to make sure it is the same, since it was not supposed to change).

This is the result we got:

Column	Data type	Range	Changed?
Patient ID	int64	1000 → 1499	No
Age	float64	0 → 195	No
Satisfaction Level	int64	1 → 5	Yes
Consultation Price	float64	50.0376... → 39,999.2201...	No
Insurance Coverage	float64	0 → 398.6022...	Yes
Consultation Duration	int64	15 → 600	No
App. Annual Income	float64	0 → 98,431	Yes

Table 7 – Updated Ranges for Numerical Columns

After analysing the ranges again, we concluded that the Satisfaction Level, the Insurance Coverage and App. Annual Income changed.

To further understand and visualize the outliers, the following box plots were drawn:

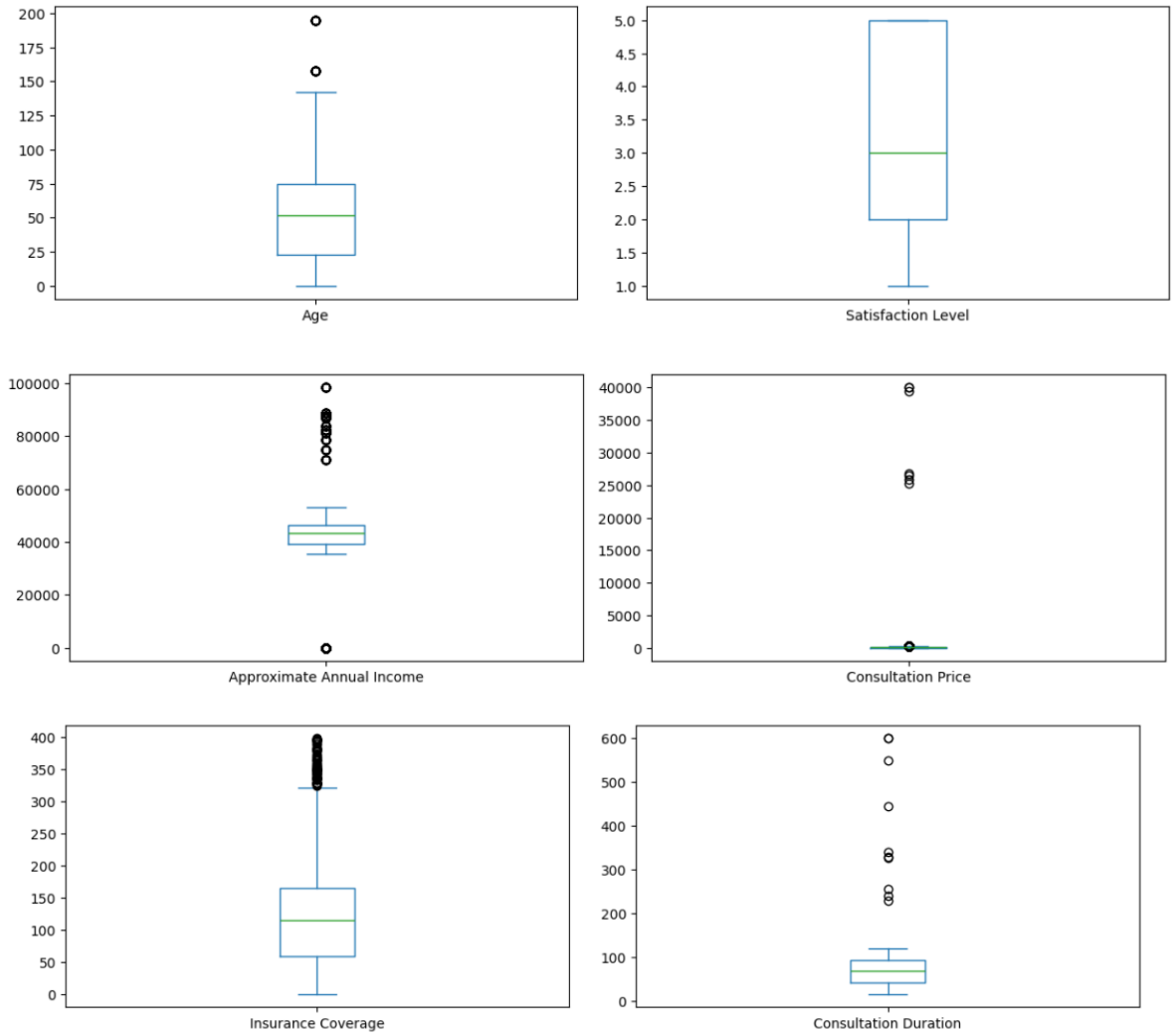


Fig. 1 – Boxplots Indicating Outliers for Numerical Features

Using the IQR we checked to see how many outliers we were dealing with in each variable:

- Age: 36;

- Satisfaction Level: 0;
- App. Annual Income: 2390;
- Consultation Price: 240;
- Insurance Coverage: 77;
- Consultation duration: 10.

Then we checked the total number of records to determine how many rows could be removed without exceeding the 5% threshold. With 10,000 rows in total, this meant we could drop no more than 500 records.

Additionally, we plotted the distribution of each variable using histograms:

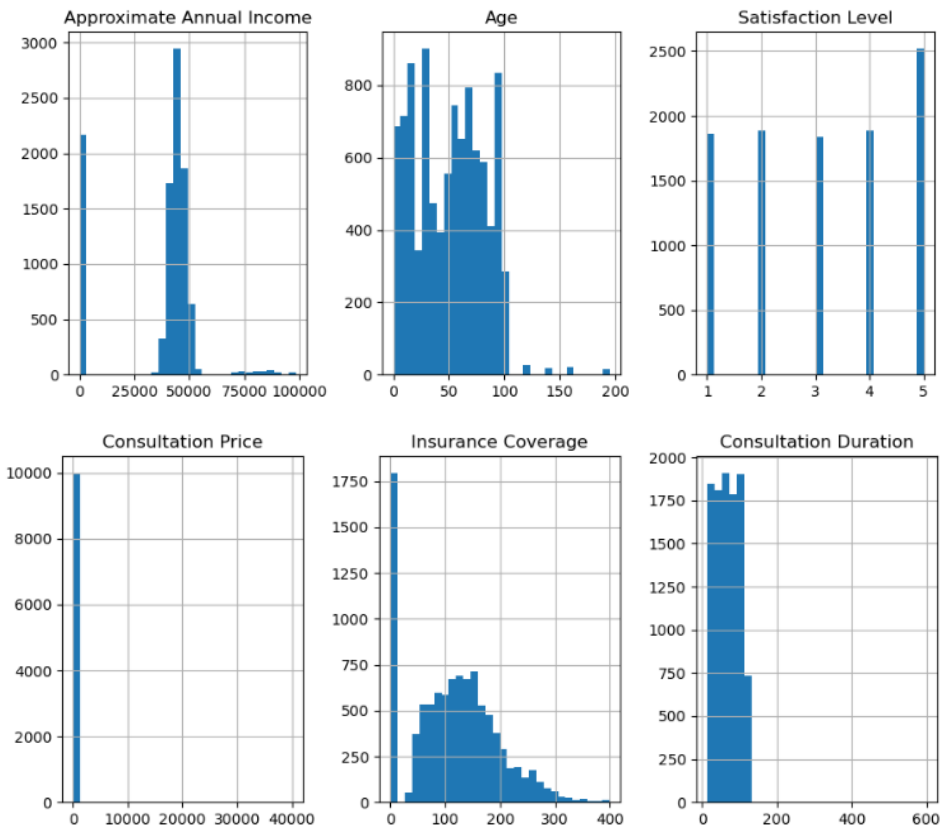


Fig. 2 – Distribution of Numerical Variables

1.6.1 Age Outliers

We first looked at unusual values in the **Age** column. **Any age above the 99th percentile was considered an outlier. There were 82 such rows**, which is only 0.82% of the dataset. These ages were clearly mistakes, and we couldn't tell if, for example, an age of 195 was meant to be 19, 95, or something else. If we kept these rows and just limited the age to 100, it would create an unnatural bunch of values at 100. So, we **decided to remove these rows** from the dataset.

1.6.2 Consultation Price Outliers

Using the same method as before, **Consultation Price was capped at the 99th percentile**, and we then checked how many rows exceeded this threshold**. There were 99 such rows**, representing only 0.99% of the dataset. These values

were clearly entry errors, and it was impossible to determine whether, for example, a consultation price of 39.999,00 should have been 39,99, 399,90, or another value. Additionally, keeping these records and simply capping them at the upper limit of 374,25 would have created artificial clustering at that boundary. Therefore, we **decided to remove these rows**.

1.6.3 Consultation Duration Outliers

Using the same method as before, **Consultation Duration was capped at the 99th percentile**, and we then checked how many rows exceeded this threshold. **There were 85 such rows**, representing only 0.85% of the dataset. These values were clearly entry errors, and it was impossible to determine whether, for example, a consultation duration of 600 should have been 60, 6, or another value. Additionally, keeping these records and simply capping them at the upper limit of 374,25 would have created artificial clustering at that boundary. Therefore, we **decided to remove these rows.**

1.6.4 Approximate Annual Income and Insurance Coverage Outliers

Although both variables contain many outliers, their ranges are reasonable and realistic. For Approximate Annual Income, as noted above, values range from 0 to around 100.000,00. Considering that this dataset includes both high-income patients (e.g., doctors and lawyers) and no-income individuals (e.g., children and students), **there is no need to remove any records**. Any decision to scale, normalize or even alter income values is left to the next team.

For Insurance Coverage, as noted above, values range from 0 to 398,6. **Considering that some patients do not have an insurance provider** (explaining the records with coverage equal to 0) and that the maximum value is realistic based on UK health insurance plans (e.g., Bupa offers coverage of up to £500 per hospital appointment or consultation), **no records were removed or adjusted**.

1.7 Handling Missing Values

Finally, missing values were assessed across the dataset. We started by checking how many non-null values each variable had and got the following:

- **Insurance Provider** had **1860** missing values;
- **Family Histor**y had **1924** missing values;
- **Education Level** had **29** missing values;
- **Approximate Annual Income** had **35** missing values;
- **Insurance Coverage** had **49** missing values;
- **Age**, **Gender**, **City of Residence**, **Profession**, **Marital Status**, **Visit Date**, **Department**, **Consultation Duration**, **Satisfaction Level**, and **Consultation Price** had **no** missing values.

1.7.1 Insurance Provider, Family History, Education Level, and Approximate Annual Income missing values

For all these columns, essentially all columns with missing values except for **"Insurance Coverage",** which was handled using a different approach, we applied **forward fill and backward fill** to impute the missing data. **Forward fill** propagates the last observed valid value forward to replace missing entries, while **backward fill** uses the next available valid value to fill in missing values. These methods were appropriate because the affected columns contain

static information for each patient, meaning the values do not change over time; therefore, we only needed a consistent way to populate the missing entries within each patient's records.

However, we also encountered cases where **all static values for a given patient were missing**, making both forward and backward fill ineffective. To address this issue, we replaced those specific cases with **"Unknown."** This approach avoids making unsupported assumptions about the data while preserving transparency and giving future teams the flexibility to apply alternative imputation strategies if needed.

1.7.2 Insurance Coverage missing values

Unlike other static patient attributes, **Insurance Coverage** varies across appointments for the same patient, as it depends on the **specific consultation, department,** and **insurance provider**. Therefore, traditional forward and backward filling were not appropriate for this variable.

Instead, we applied a **group-based imputation strategy**. The dataset was grouped by **Insurance Provider** and **Department**, ensuring that missing values were imputed using information from consultations with the same clinical and billing context. Within each group, missing **Insurance Coverage** values were replaced with the median coverage value observed in that group.

The **median was chosen over the mean** because, as observed in the data visualization performed earlier, the distribution of Insurance Coverage is **right-skewed** and contains many zero values, corresponding to patients who had no insurance coverage for specific consultations. Even after treating outliers, this distributional asymmetry makes the median a **more robust and representative measure of central tendency,** particularly within smaller provider, department subgroups.

This approach preserves the **pricing structure and coverage behaviour** specific to each provider and department, while avoiding cross-patient leakage and maintaining statistical consistency within the dataset.

Conclusion of Part A

At the end of Part A, the dataset was fully cleaned and standardized. Duplicate records were removed, categorical and numerical inconsistencies were corrected, illogical values were filtered, missing values were handled appropriately, and outliers were treated in an appropriate manner, stabilizing the numerical distributions.

2. Part B - Feature Engineering & Analytic Base Table (ABT) Construction

The objective of Part B was to transform the cleaned visit-level dataset obtained in Part A into a **patient-level Analytic Base Table (ABT)** suitable for descriptive analysis, clustering, and future predictive modelling. Since the raw data contained multiple consultation records per patient, feature engineering and aggregation were required to represent each patient with a single summarized record.

We began building the ABT by grouping the finalized, cleaned dataset by patient ID so that each row represents a single patient. For demographic variables (Age, Gender, City of Residence, Profession, Education Level, Marital Status, Family History, Approximate Annual Income, and Insurance Provider) we retained the first value, as these attributes remain constant for each patient. For the variables Insurance Coverage, Satisfaction Level, Consultation Duration, and Consultation Price, we calculated their mean values and renamed them using the prefix 'Avg_'followed by the original column name.

Next, we created an **'Age Group'** column that categorizes each patient based on their age using a custom function. Similarly, we generated an **'Income Group'** column to classify patients according to their income bracket. We then added a **'Most Visited Department'** column to identify each patient's most frequently visited department.

To capture recency, we introduced a **'Last Visit (Days)'** column indicating how many days have passed since the patient's most recent hospital visit. We also created two additional columns: **'Total Spent'**, representing the total amount a patient spent across all visits, and **'Satisfaction Volatility'**, measuring the variation in satisfaction scores using the standard deviation (with a value of 0 for patients with only one record).

We then added department-specific visit count columns showing how many times each patient visited each department, followed by a column representing the **total insurance coverage** a patient received. Using total spending and total coverage, we calculated the **'Insurance Coverage Ratio'**, which expresses the proportion of costs covered by insurance (ranging from 0 to 1, where 1 indicates full coverage).

After that, we included **'First Visit'** and **'Last Visit'** columns to record each patient's earliest and most recent visit dates. Finally, we created an **'Average Insurance Coverage per Visit'** column to show how much insurance covered on average for each patient's visits.

While multiple features were engineered, the following were identified as the most critical for our subsequent analysis:

- **Satisfaction Volatility:** Essential for identifying inconsistent patient experiences, which we hypothesized to be a stronger driver of churn than average satisfaction alone.
- **Total Spent:** A key metric for patient value segmentation, allowing us to move beyond simple visit counts.
- **Insurance Coverage Ratio:** Critical for analysing financial friction points, distinguishing between fully covered patients and those paying out-of-pocket.
- **Last Visit (Days):** The primary feature used to measure recency and analyse potential churn risk.

Finally, we performed several safety checks to ensure data integrity. We verified that all patient records were unique and that no duplicates were present. We also checked for any missing values that might have been overlooked and reviewed the data types of all columns to confirm consistency. No issues were found, and the ABT was successfully completed.

These are all the columns present in the ABT:

Patient ID	Age	City of Residence	Profession	Education level
Marital Status	Family History	Approximate Annual Income	Insurance Provider	Avg_Insurance_Coverage
Avg_Satisfaction_Level	Avg_Consultation_Duration	Avg_Consultation_Price	Age Group	Income Group
Total Visits	Most Visited Department	Last Visit (Days)	Total Spent	Satisfaction Volatility
Visits_Allergology	Visits_Cardiology	Visits_Dermatology	Visits_ENT	Visits_Emergency
Visits_Endocrinology	Visits_Gastroenterology	Visits_General Practice	Visits_Neurology	Visits_Orthopedics
Visits_Psychiatry	Visits_Pulmonology	Visits_Rheumatology	Total Insurance Coverage	Insurance Coverage Ratio
First Visit Date	Last Visit Date			

Previously existing columns

Newly added columns

3. Part C – Exploratory Visualization & Insights

The objective of Part C was to perform an **exploratory visual analysis** of the patient-level Analytic Base Table (ABT) constructed in Part B. The purpose of this phase was to uncover key patterns related to **patient demographics, service usage, satisfaction, and financial behaviour**, and to provide **initial data-driven insights** to support City Hospital's management and the next analytics team.

A series of visualizations was created to explore different dimensions of patient behaviour and hospital performance.

3.1 Patient demographics: Age, Gender, and Income distributions

We began by visualizing the distributions of patient demographics, including **Age, Approximate Annual Income, Gender, City of Residence, Profession, Education Level, Marital Status, Family History, and Insurance Provider**. Among these, **Age** and **Approximate Annual Income** are numerical variables. The resulting distributions are as follows:

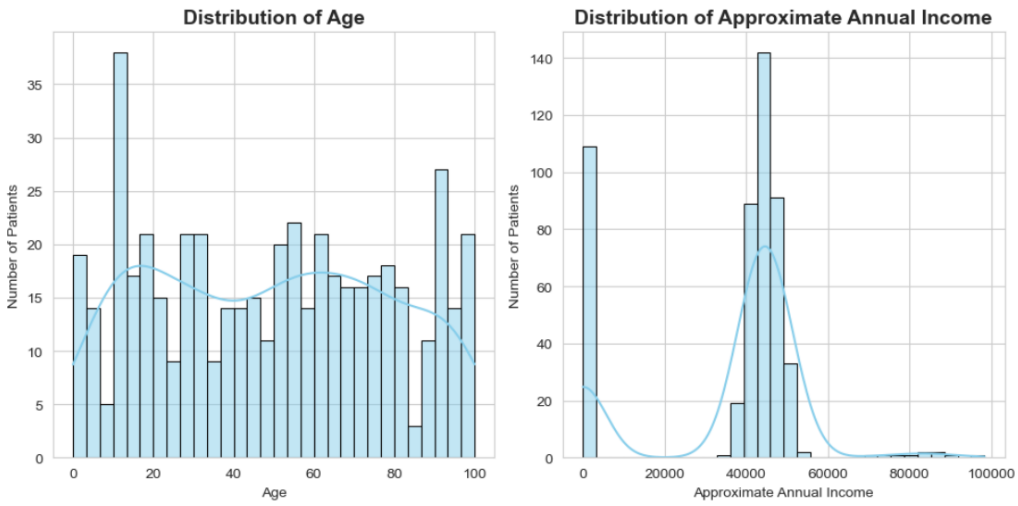
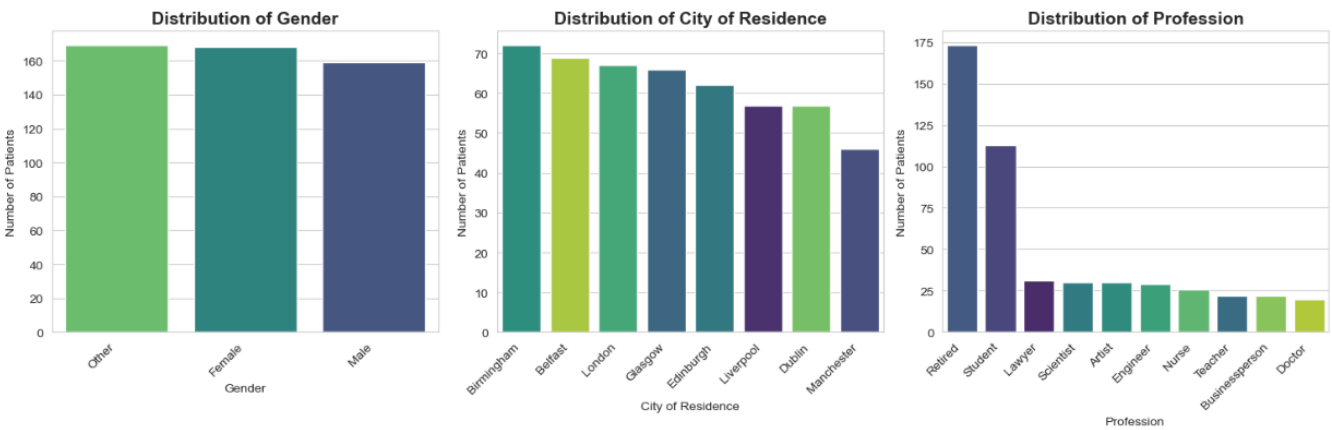


Fig. 3 – Distribution of Numerical Variables



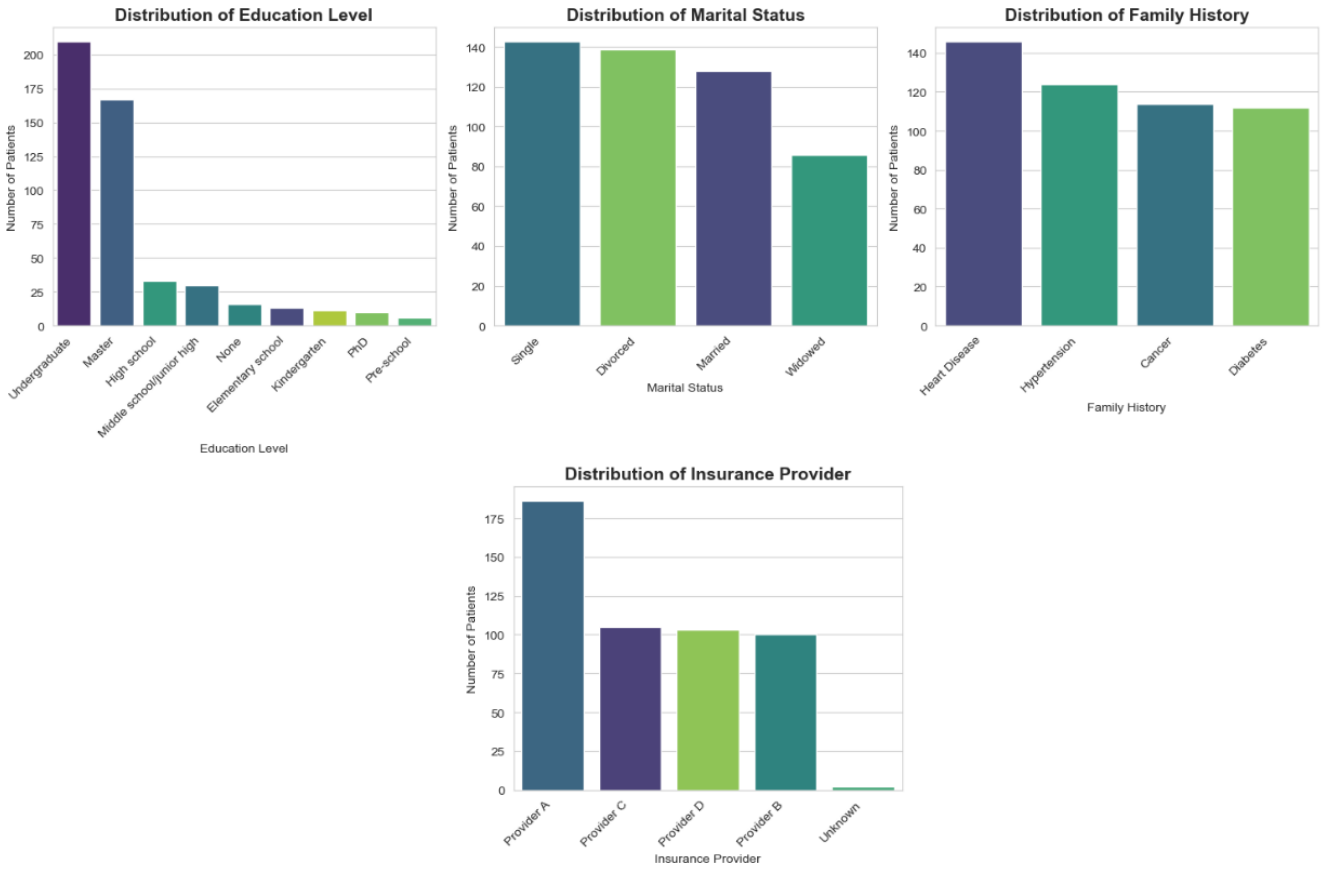


Fig. 4 – Distribution of Categorical Variables

These visualizations allowed us to examine the distributions of numerical variables and identify the most frequent categories among the categorical variables.

Key insights include:

- **Age** does not follow a specific distribution;
- **Approximate Annual Income**, after removing extreme values (0 and incomes above 60K), follows an approximately normal distribution;
- **Patients identifying as "Other" gender** are the most frequent visitors, though male and female patients have similar counts;
- **Patients from Birmingham** represent the largest group of visitors;
- **Retired patients** are the most frequent visitors by profession;
- **Undergraduate education** is the most common education level among patients;
- **Most patients are single;**
- **A majority of patients have a family history of heart disease;**
- **Provider A** is the most common insurance provider.

3.2 Visitation related visualizations: Consultation Duration, Consultation Price, Satisfaction Level, and Insurance Coverage distributions

Next, we decided to plot the distributions for the variables that change between every visit. These are Consultation Duration, Consultation Price, Satisfaction Level, and Insurance Coverage. This is what we got:

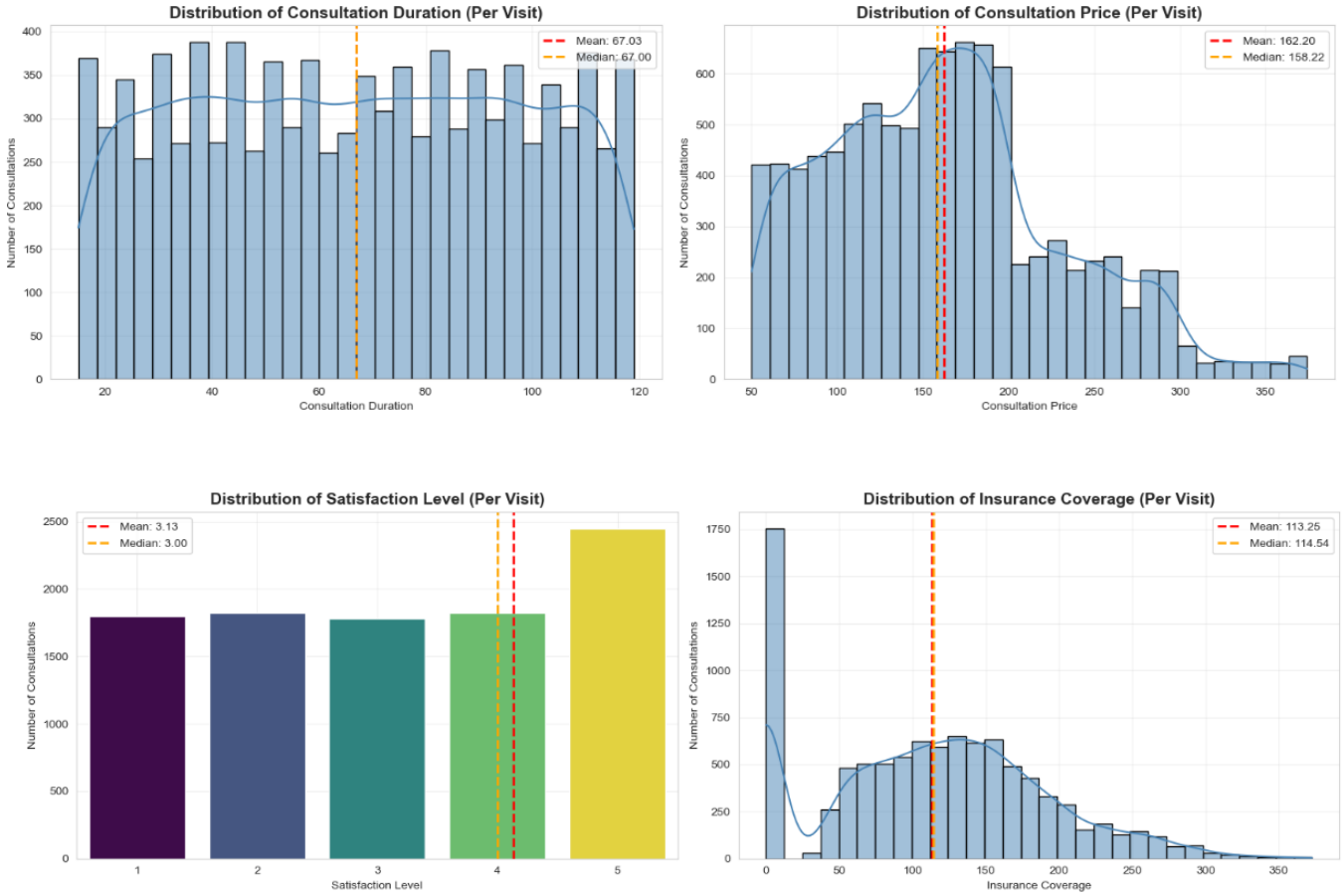


Fig. 5 – Distribution of consultation related variables

After plotting the visualizations, several clear insights emerged:

- **Consultation Duration:** The distribution is nearly uniform, with the mean and median matching closely, indicating a symmetric shape and no significant skew;
- **Consultation Price:** Most prices fall between 120 and 200, but higher values push the mean above the median, revealing a right-skewed distribution with notable variability;
- **Satisfaction Level:** The most common rating given was 5, yet the average is 3, showing that although many visits were highly rated, a considerable number received very low satisfaction scores;
- **Insurance Coverage:** A large share of visits receives little to no coverage, while the remainder cluster between 60 and 150. The similar mean and median suggest balance in the bulk of the data, though a long right tail indicates the presence of high-coverage outliers.

3.3 Department & Service usage visualization

To better understand and gain insights about the departments and their usage not only by age but also by gender, we decided to plot the following graphs:

Fig. 6 – Department & Service usage visualization

After plotting the visualizations, several clear insights emerged:

- **Total Visits per Department:** The two departments with the highest number of visits were General Practice and Psychiatry. All remaining departments had noticeably fewer visits and showed similar visit patterns to one another;
- **Total Visits Over Time (Monthly):** Across the period from January to June, May registered the highest volume of visits, with March and January also showing strong activity;
- **Department Usage by Gender:** Across males, females, and other genders, Psychiatry consistently emerged as the most visited department;
- **Department Usage by Age Group:** Seniors (65+) were the most frequent users across all departments, particularly in General Practice and Psychiatry. Minors (0–17) and mature adults (50–64) followed the same

overall pattern. Middle-aged adults (35–49) and young adults (18–34) displayed a more even distribution of visits across departments, except for noticeable increases in General Practice and Psychiatry.

3.4 Satisfaction Related Visualizations

It was also very important to visualize the overall satisfaction levels of the patients and understand how it varies across different groups, namely how the satisfaction levels change by Income Level, by Department, by age group, etc. To better understand this, these are the graphs we plotted:

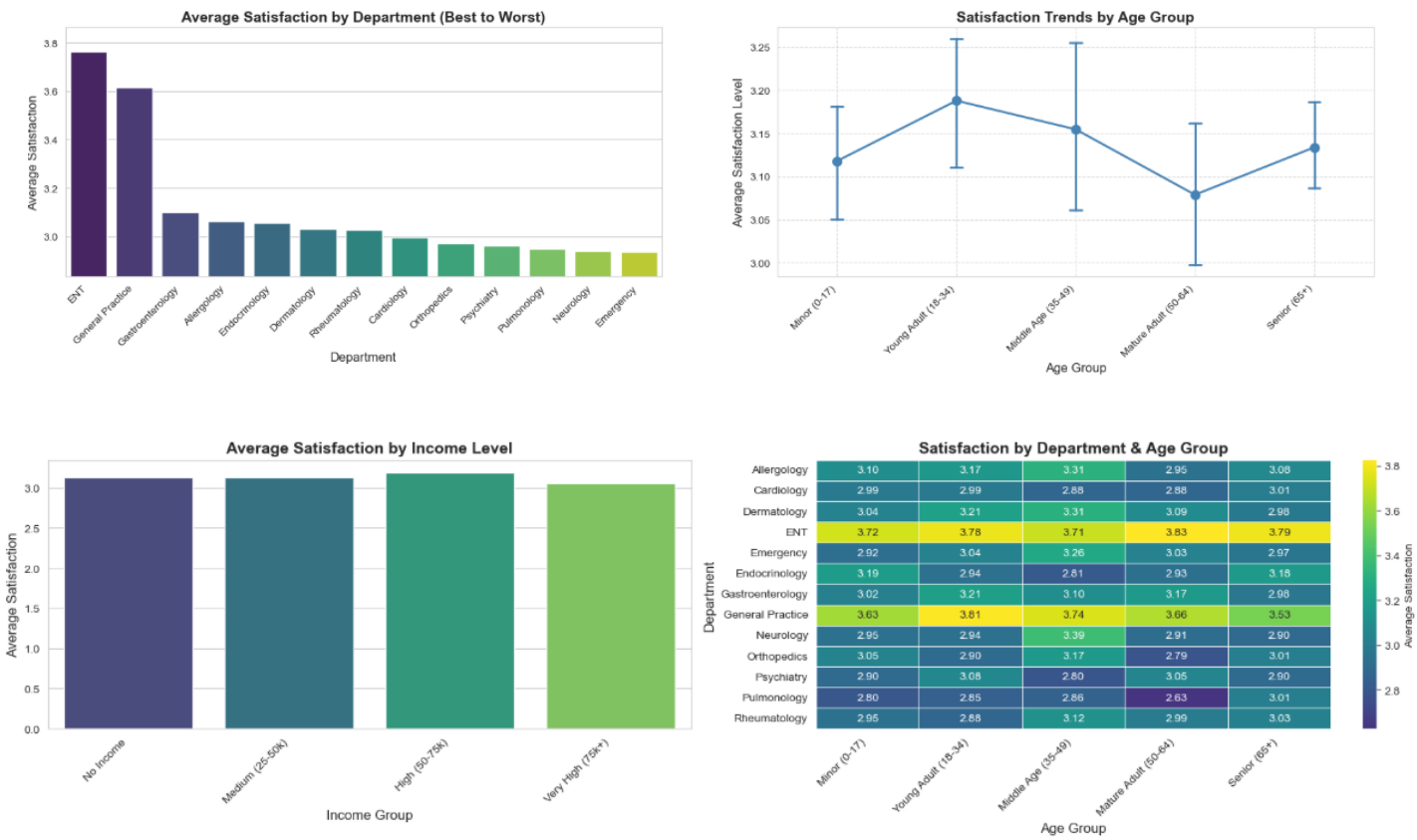


Fig. 7 – Satisfaction related visualizations

These are the main key insights we extracted from each graph:

- **Average Satisfaction by Department:** ENT and General Practice stand out with the highest average satisfaction levels. All other departments show a substantial drop in comparison. Gastroenterology, Allergology, Endocrinology, Dermatology, and Rheumatology maintain averages slightly above three, while the remaining departments sit just below three;
- **Satisfaction Trends by Age Group:** Young Adults and Middle-Aged Adults report the highest satisfaction levels, whereas Mature Adults show the lowest. Still, overall differences across age groups remain relatively small;

- **Average Satisfaction by Income Level:** All income groups exhibit nearly identical satisfaction averages, indicating that annual income does not influence satisfaction—an observation later confirmed by the correlation matrix;
- **Satisfaction by Department and Age Group:** The department is the strongest determinant of satisfaction, with ENT and General Practice consistently achieving top ratings, while Pulmonology and Cardiology receive the lowest. Additionally, high consultation prices—typically within the 160–190 range but extending far higher, do not necessarily result in higher satisfaction. In fact, the most highly rated services tend to be lower-cost, high-volume general care. The 200 price point appears to act as a threshold where visit volume drops sharply, suggesting strong cost sensitivity despite the wide variation in satisfaction across specialties.

3.5 Additional visualizations

To add extra insights and explore even further the distributions and the relation between variables we decided to plot new graphs.

3.5.1 Correlation Matrix of ABT Features

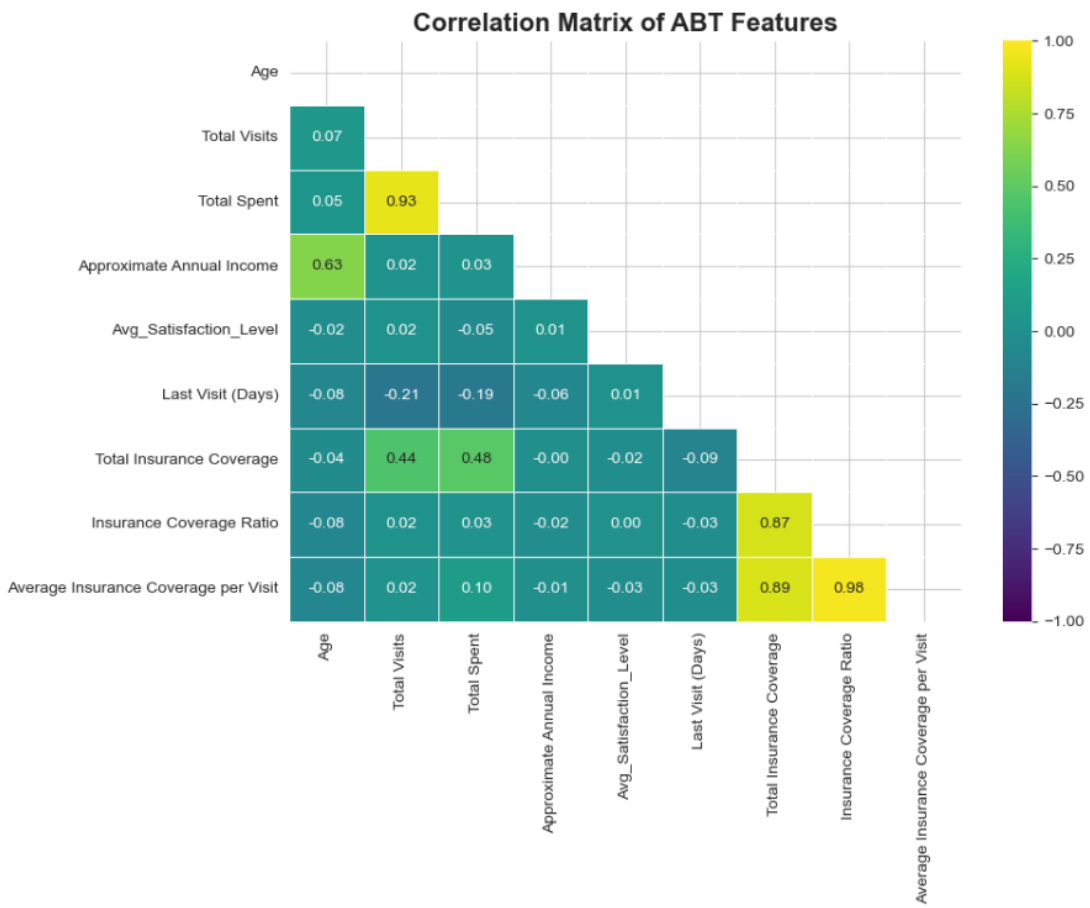


Fig. 8 – Correlation Matrix of ABT Features

Major Key Insights

1. The most important insight comes from what is missing. The row for Avg_Satisfaction_Level is entirely teal, with correlation values near zero (ranging from -0.05 to 0.01).

- Patient satisfaction has **zero correlation** with how much money they spend, their income, their age, or their insurance coverage;
- You cannot "buy" satisfaction with expensive treatments, nor are wealthier patients naturally happier. This strongly validates the previous heatmap's finding that satisfaction is driven by **operational performance** (e.g., Department efficiency) rather than patient demographics.

2. There is a notable positive correlation of 0.63 between Age and Approximate Annual Income.

- As your patients get older, their income significantly increases;
- Your older demographic (who, recalling previous data, are the "grumpier" ones) are also your wealthiest. If they are unhappy (as seen in Pulmonology/Rheumatology), you are at risk of alienating your most financially capable customer base.

3. Spend is Driven Purely by Volume

There is an extremely strong correlation of 0.93 between Total Visits and Total Spent.

- Revenue is almost perfectly linear with visit frequency. There are likely no major "upsell" outliers where a single visit generates massive revenue; rather, high-value patients are simply those who show up often.

4. Insurance Redundancy

The bright yellow block in the bottom right shows very high correlations (0.87-0.98) between Total Insurance Coverage, Insurance Coverage Ratio, and Average Insurance Coverage per Visit.

- These three metrics are statistically redundant (collinear). In future predictive modelling, the next team will likely only need one of these variables to avoid skewing the data. The choice of which ones to drop is left to the next team, so they can decide which one fits their model better.

3.5.2 Patient Value Analysis: Spend vs. Frequency

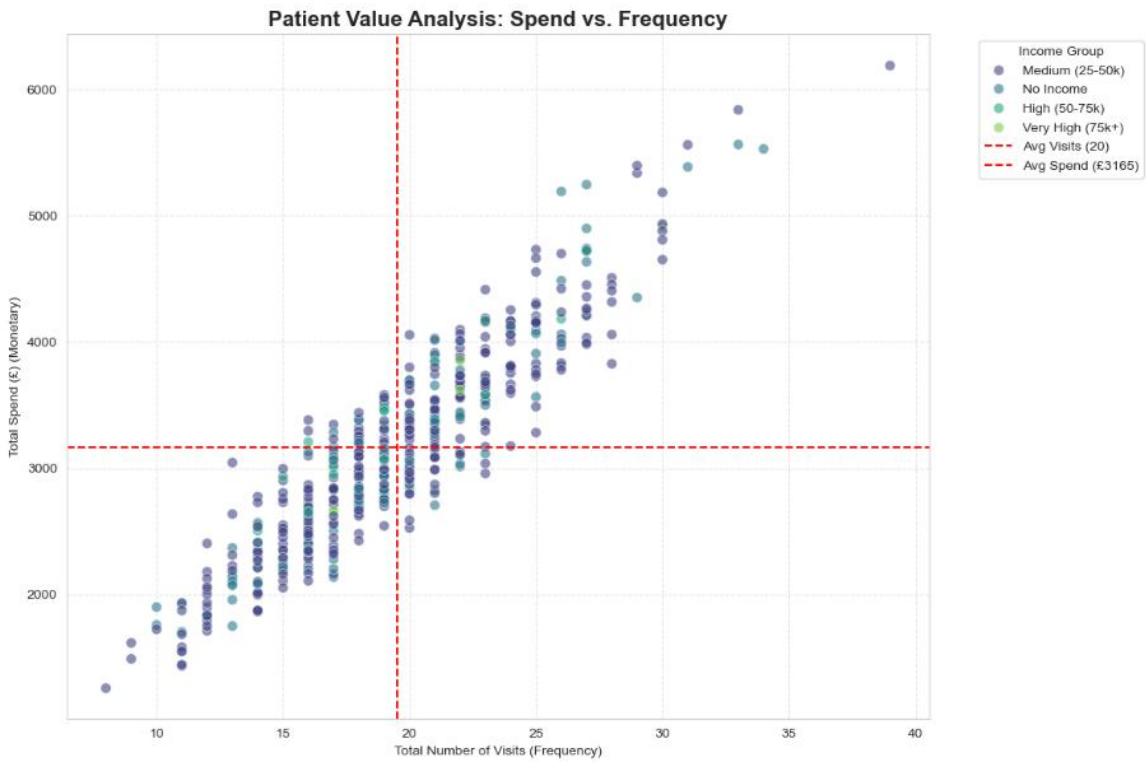


Fig. 9 – Patient Value Analysis: Spend vs Frequency

There is a strict, positive linear relationship between the number of visits and the total amount spent. This confirms that revenue growth is driven almost entirely by volume (getting patients to come back) rather than by increasing the cost per visit for specific high-net-worth individuals.

Key Insights

1. Volume is key

- The data points form a tight, diagonal line extending from bottom-left to top-right;
- Generally, there are no patients who visit once but spend thousands (no high-value, low-frequency anomalies). Every pound of revenue is earned through repeated visits.

2. The "VIP" Quadrant (Target Segment)

- The red dashed lines create a "crosshair" at the averages: 20 visits and £3,165;
- Patients in the top-right quadrant (above 20 visits, spending >£3,165) are our "VIPs." These are our most loyal customers;
- Patients in the bottom-left are below average. So, the strategy for the Hospital would be trying to retain them, bumping their amount of visits from 10, for example, to 20, in order to increase their values.

3. Income Does Not Dictate Hospital Spend

- The "Very High Income" (light green) and "Medium Income" (purple) patients are scattered relatively evenly across the entire line;
- There are "No Income" patients (teal dots) in the top-right quadrant spending over £5,000, and "High Income" patients in the bottom-left spending less than £2,000;
- A patient's wealth (outside the hospital) does not predict their value to the hospital. Sickness is the equalizer. A wealthy healthy person is worth less to the hospital than a lower-income patient with a chronic condition requiring 30+ visits.

4. No "Whales"

- In many businesses, you look for "whales"—customers who spend disproportionately more than others;
- Here, the cap is visible. The highest frequency is just under 40 visits, and the highest spend is roughly £6,000. There is no one spending £20,000. This suggests the revenue is capped by clinical necessity (there are only so many times you *need* to see a doctor).

3.5.3 Average Spending Breakdown by Insurance Provider

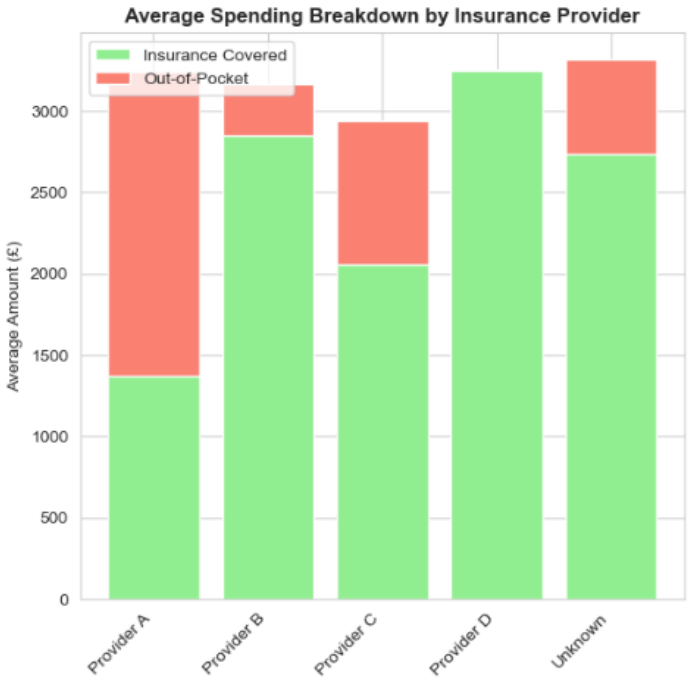


Fig. 10 – Average Spending Breakdown by insurance Provider

This graph shows the proportion of total medical expenses covered by each insurance provider. The key insight is that patients with **Provider D** had their medical costs fully covered by insurance. In contrast, **Provider A** covers, on average, less than 50% of each patient's total expenses. **Providers B and C** both cover, on average, more than 50% of patients' total medical costs.

3.5.4 Satisfaction Consistency vs Visit Frequency and Departments with most Inconsistent Satisfaction

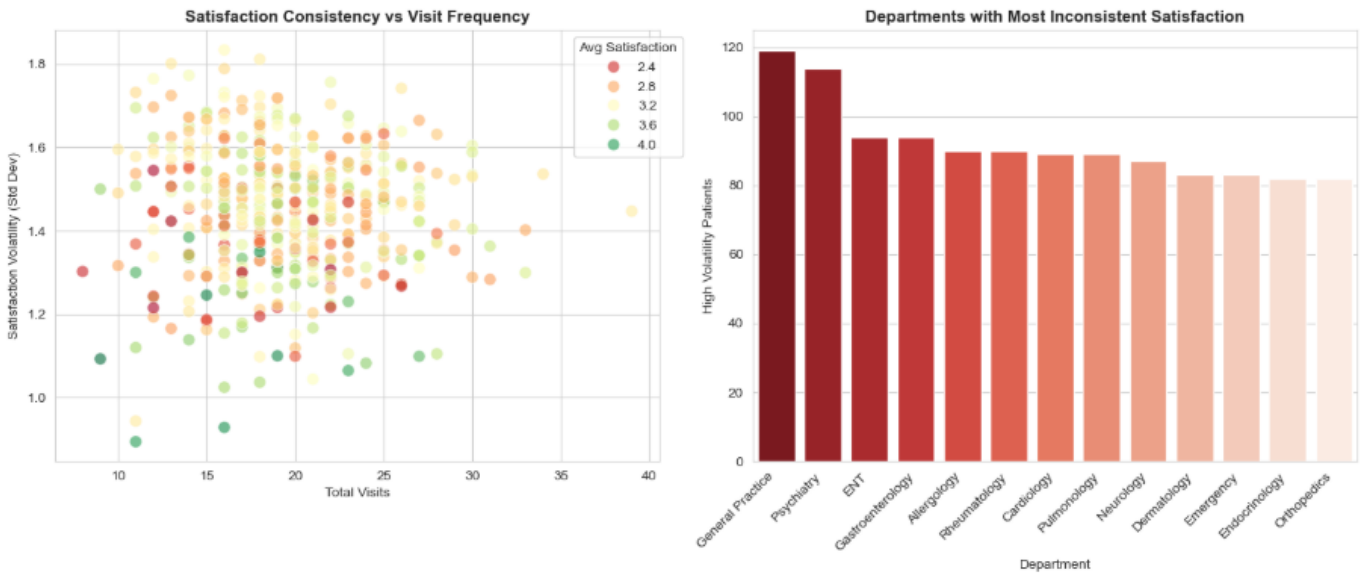


Fig. 11 – Satisfaction Consistency vs Visit Frequency and Departments with most Inconsistent Satisfaction

First Graph:

1. The "Captive Audience" Problem (High Volume, Mediocre Experience)

Looking at the far right of the chart (patients with **25+ visits**):

- There are almost **no deep green dots** (high satisfaction) in this high-frequency zone;
- Most of these loyal patients are yellow or light orange, indicating average to below-average satisfaction;
- The most valuable customers (revenue-wise) are likely "trapped" by chronic conditions. They keep coming back because they *have* to, not because they enjoy the service. They are enduring a mediocre experience.

2. Consistency Drives True Satisfaction

- The **deep green dots** (Highest Satisfaction, ~4.0) are concentrated in the lower-left to middle areas;
- Crucially, many of these green dots sit lower on the Y-axis (lower volatility);
- Patients rate highest when the hospital is **boring but reliable**. A patient who has three "good" visits is happier than a patient who has one "amazing" visit and one "terrible" one.

3. The "Rollercoaster" Effect of Long-Term Care

- As the number of visits increases (moving right along the X-axis), the "Volatility" (Y-axis) generally stays high, hovering between **1.2 and 1.6**;

- It is operationally difficult to maintain a perfect experience over 30+ visits. Eventually, a long-term patient hits a bad day—a rude receptionist, a long wait time, or a billing error. This "regression to the mean" prevents long-term patients from remaining "Super Promoters."

4. The "Red Alert" Cluster

- There is a scattering of **red/dark orange dots** (Low Satisfaction < 2.8) throughout the 10–25 visit range;
- These are patients who visit frequently and consistently have bad experiences (or highly volatile ones that average out to bad);
- These are the biggest liability. They interact with the hospital often and likely share their negative experiences with others (Word of Mouth), yet they are still using hospital resources.

Second Graph

The second graph, “Departments with Most Inconsistent Satisfaction,” displays the number of high-volatility patients per department. The purpose of this graph is to highlight the departments most affected by patients with highly variable satisfaction, helping to explain overall satisfaction patterns within each department.

A high-volatility patient is defined as one who falls into the fourth quartile (0.75 and above) of the *Satisfaction Volatility* metric we created. These patients exhibit the greatest fluctuations in their satisfaction scores and therefore have a disproportionate impact on departmental averages.

From the data, General Practice and Psychiatry stand out, each with over 110 high-volatility patients. This indicates that these departments experience the highest inconsistency in patient satisfaction. The remaining departments all have between 80 and 90 high-volatility patients, suggesting a similar level of variability among their patient populations.

Departments with a higher concentration of high-volatility patients may benefit most from targeted operational improvements, personalized follow-ups, or enhanced patient experience initiatives, as these actions could significantly stabilize overall satisfaction scores.

3.5.5 Patient Recency Distribution over the 6-month period

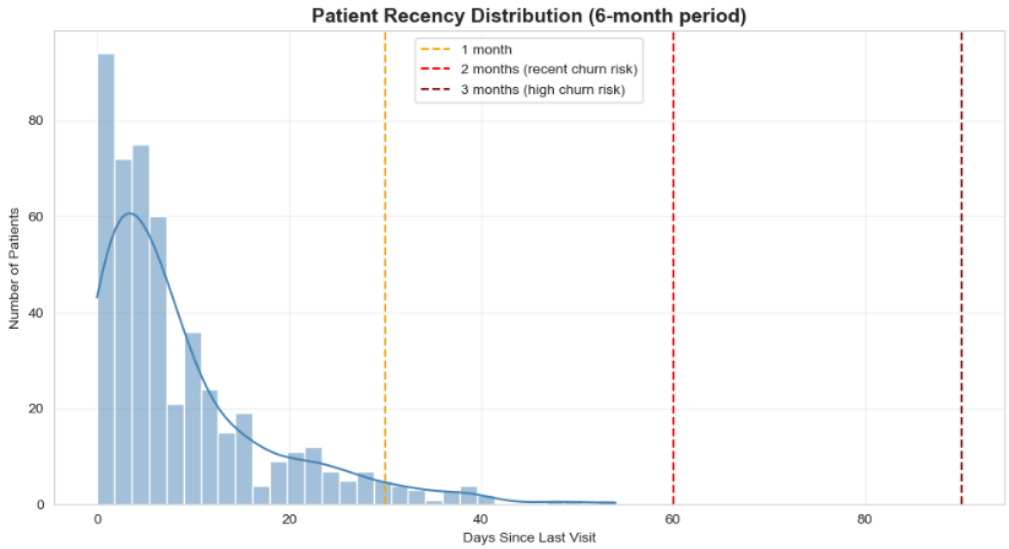


Fig. 12 – Patient Recency Distribution (6-month Period)

Our patient base is extremely "active." The vast majority of patients have visited the facility within the last 30 days. There is virtually no "churn risk" visible in this specific 6-month snapshot, as almost no patients have gone more than 60 days without a visit.

Key Insights

1. Extreme Engagement (The "0-10 Day" Spike)

- The tallest bars are clustered at the very beginning, between **0 and 10 day;**
- The peak volume is roughly **95 patients** who visited just a few days ago;
- This suggests a high-frequency operational model (e.g., physical therapy, dialysis, or post-op recovery) rather than annual check-ups. Patients are currently "in the cycle" of care.

2. The Drop-Off at 30 Days

- There is a significant natural decay in visits as time goes on. By the time we reach the **30-day mark (Orange Line)**, the number of patients drops to single digits;
- If a patient hasn't come back in a month, they are statistical outliers in this specific dataset. A "normal" patient interval here is essentially weekly or bi-weekly.

3. "Churn Risk" is Theoretical, Not Actual

- Churn risk is the probability that a patient stops using the service;
- The chart includes red dashed lines for **2 months (60 days)** and **3 months (90 days)** to indicate churn risk;

- However, the data (blue bars and line) flatlines near zero long before reaching the 60-day mark;
- In this specific 6-month window, there are almost no "dormant" patients. Everyone is either currently active or has "completed" their care cycle very recently.

4. Operational Implication

- Since most patients visited in the last 2 weeks, the front-desk and clinical staff are seeing the *same patients* constantly. This reinforces the earlier finding about "Visit Frequency";
- If these patients are experiencing the "volatility" in satisfaction mentioned earlier, they are experiencing it *right now*. There is no "cooling off" period.

Synthesis with Previous Data

- There are patients visiting weekly (high frequency) who are likely experiencing the inconsistent satisfaction levels we saw in the scatter plot;
- Since revenue is driven by volume, the cash flow depends entirely on this short-term intensity. If this 0–30 day cycle breaks, revenue stops immediately because there is no "long tail" of patients returning after 3 months.

4. Suggestions for the next analytics team

4.1 Operational Targets & Metric Focus

The correlation matrix and scatter plots demonstrated that satisfaction is independent of financial metrics (income, total spend) and purely driven by the patient experience.

- Primary Target: Stabilize "High-Frequency" Satisfaction.
  - **Goal:** Reduce "Satisfaction Volatility" for patients with >20 visits;
  - **Why:** These patients generate the most revenue but currently experience the most inconsistent service. A churn event here is costly;
  - **Actionable Metric:** Track standard deviation of satisfaction *per patient* over time, not just average satisfaction.
- Departmental overhaul for Pulmonology & Cardiology.
  - **Goal:** Lift average satisfaction scores in these departments to a baseline of 3.0;
  - **Why:** These departments are consistently underperforming across *all* age groups,
  - **Hypothesis to Test:** Investigate if the low scores are due to wait times, as these departments likely have complex cases that cause scheduling delays.

4.2 Candidate Segmentation Models

The standard "Gold/Silver/Bronze" segmentation based on spend is insufficient because spend is purely linear with visits. Instead, we propose behavioral segmentation:

- **Segment A: "The Captive Loyals" (High Frequency / High Volatility)**
  - **Profile:** 20+ visits, high total spend, but mixed satisfaction scores;
  - **Strategy:** "Concierge" service. Prioritize their scheduling to reduce friction. They are already loyal by necessity; the goal is to prevent burnout and negative word-of-mouth.
- **Segment B: "The One-Month Wonders" (High Recency / Short Cycle)**
  - **Profile:** Patients with 5-10 visits clustered entirely within a 30-day window;
  - **Strategy:** Efficiency. These are likely post-op or acute care cycles. The goal is rapid, frictionless processing (fast check-in/out) as they are visiting very frequently in a short burst.
- **Segment C: "The Price-Sensitive Prospects"**
  - **Profile:** Patients dropping off after consultation prices exceed **£200**;
  - **Strategy:** Psychological Pricing. Test pricing bundles or payment plans for treatments crossing this threshold to smooth the volume drop-off observed in the price histogram.

4.3 Predictive Modeling Recommendations

- Feature Selection:
  - **DROP:** Total Insurance Coverage and Average Insurance Coverage per Visit. They are 98% correlated with Insurance Coverage Ratio. Keeping them will cause multicollinearity issues in regression models;
  - **KEEP:** Satisfaction Volatility. This derived feature is a stronger predictor of potential churn than Avg Satisfaction Level because it captures the *risk* of a bad experience.
- Churn Prediction:
  - **Refinement:** The current 6-month window shows almost no "long-term" churn because most patients are active;
  - **Recommendation:** The next team should request historical data spanning 12-24 months to properly train a churn model. With only 6 months, you are only seeing active care cycles, not the "lost" patients.

4.4 Immediate "Quick Wins"

- **Investigate the "Middle Age" Neurology Anomaly:** Patients aged 35-49 rate Neurology significantly higher (3.39) than any other age group. Identify *what* is working for this demographic (e.g., specific doctors, treatment types) and replicate it for older patients;
- **Audit "Provider D":** This insurance provider covers 100% of costs. Investigate if this generous coverage correlates with higher utilization rates (moral hazard) compared to "Provider A" patients who pay 50% out-of-pocket.